

## Question 1

### (i) Number of Comparisons:

On the first pass bubble sort compares the first element with all other elements in the list, doing  $n-1$  comparisons. On the second pass it compares the first element with all other elements minus the last element at the end of the array, doing  $n-2$  comparisons. This is repeated until there is only the first element remaining. Therefore the number of comparisons is equal to:

$$\text{Total Comparisons} = (n-1) + (n-2) + \dots + 1$$

Which is the sum of the first  $n-1$  integers and is given by the arithmetic formula said to be derived by baby Gauss.

$$\text{Total Comparisons} = \frac{(n-1) \times n}{2}$$

Which has a complexity analysis of  $O(n^2)$ .

### (ii) Average-Case Number of Swaps:

A swap only occurs when an element is larger than the element directly behind it which means taking the average should only happen around half the time. Therefore the derivation is very straightforward, since we know bubble sort performs a comparison every time it passes an element, there is a chance that element must be swapped, so the average case number of swaps is just half the number of comparisons bubble sort performs:

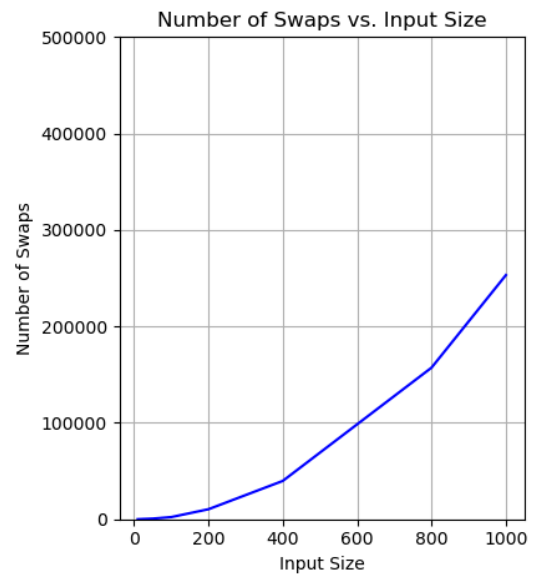
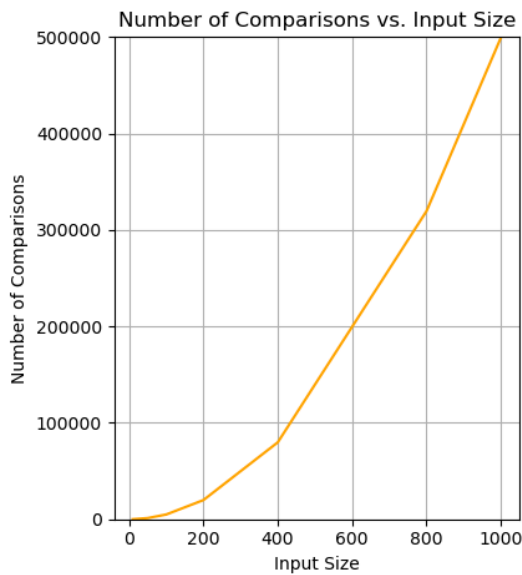
$$\text{Average-Case Number of Swaps} = \text{Total Comparisons} / 2$$

$$\text{Average-Case Number of Swaps} = \frac{(n-1) \times n}{2} \times \frac{1}{2}$$

$$\text{Average-Case Number of Swaps} = \frac{(n-1) \times n}{4}$$

Which also has a complexity analysis of  $O(n^2)$ .

#### Question 4



Both graphs following a quadratic path which matches our analysis above where we proved both the number of comparisons and average case number of swaps exhibit a quadratic relationship with the size of the input array. The graph of swaps vs input size is also exactly half of the number of comparisons vs input size which makes perfect sense and matches our derived formulas where average case swaps were really just total comparisons divided by 2.